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Assessment 2

Review Learning Objectives

coach

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Assignment details

Due
Dec 18, 11:59 PM IST

Attempts
Unlimited

Retry

Submitted
Dec 12, 3:07 AM IST

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1. Which of the following steps are involved in creating and executing the Deutsch algorithm circuit in Qiskit? **1 / 1 point**

☑ Apply the Hadamard gate to the qubits.

Correct
Correct! The Hadamard gate is used to create superposition, which is essential for the Deutsch algorithm.

☑ Initialize the qubits to the $|0\rangle$ state.

Correct
Correct! Initializing the qubits is the first step in setting up the Deutsch algorithm circuit.

☑ Measure the qubits to obtain the result.

Correct
Correct! Measuring the qubits at the end of the computation provides the result of the Deutsch algorithm.

☐ Apply the Toffoli gate to entangle the qubits.

☐ Execute the circuit on a classical simulator.

2. Which of the following statements correctly describe the basics and potential applications of Grover's search algorithm? **0.6 / 1 point**

☐ One potential application of Grover's algorithm is in cryptographic key search.

☑ Grover's algorithm can be used for solving systems of linear equations.

This should not be selected
Incorrect. Solving systems of linear equations is not a typical application of Grover's algorithm.

☐ Grover's algorithm can be used to factor large numbers efficiently.

☐ Grover's algorithm can be implemented on classical computers for similar speedup.

☑ Grover's algorithm provides a quadratic speedup for searching unsorted databases.

Correct
Correct! This is one of the key advantages of Grover's algorithm.

3. What are some key steps in implementing Grover's search algorithm using Qiskit? **1 / 1 point**

☐ Measure the quantum state before applying the Grover operator.

☑ Create a superposition of all possible states.

Correct
Correct! Creating a superposition is essential for the quantum parallelism in Grover's algorithm.

☑ Define the oracle for marking the correct state.

Correct
Correct! Defining the oracle is a crucial step in Grover's search algorithm.

☑ Apply the Grover operator multiple times.

Correct
Correct! Repeatedly applying the Grover operator amplifies the probability of measuring the correct state.

☐ Initialize the qubits in the $|1\rangle$ state.

4. Which of the following gates is essential for modular exponentiation in Shor's algorithm? **1 / 1 point**

○ CNOT gate

● Uf gate

○ Hadamard gate

○ Pauli-X gate

Correct
Correct! The Uf gate is used for modular exponentiation in Shor's algorithm.

5. What is the primary significance of Shor's algorithm in quantum computing? **1 / 1 point**

○ It enables quantum computers to solve differential equations directly.

○ It provides an exponential speedup for searching unsorted databases.

● It can efficiently factorize large numbers, impacting encryption methods.

○ It can solve NP-complete problems in polynomial time.

Correct
Correct! Shor's algorithm can efficiently factorize large numbers, which has significant implications for encryption.